

Shengzhe Li

☎ (+86) 15111593537 | ✉ Email | in LinkedIn | 🏠 Personal Page

EDUCATION

University of Amsterdam

M.S. in Physics (in progress)

Amsterdam, The Netherlands

Sep. 2024 – Present

- GPA: 8.06 / 10.00
- *Coursework*: Quantum Field Theory I-III(QFT), AQFT, Condensed Matter Theory(CMT), ACMT, Advanced numerical methods in many-body physics, Group Theory, Mathematical Methods in Theoretical Physics, Non-equilibrium Statistical Mechanics, Topological Materials, Correlated Quantum Matter.

Sichuan University

B.S. in Physics

Chengdu, China

Sep. 2021 – Jun. 2024

- GPA: 3.75/4.0 Average: 88.95/100 Rank: 14/103.
- *Coursework*: Quantum Mechanics, Electrodynamics, Statistical Physics, Computational Physics, Classical Mechanics.

National Tsing Hua University

Exchange Student, Department of Physics

Hsinchu

Sep. 2023 – Jan. 2024

RESEARCH EXPERIENCE

Manipulating Exciton with Light

Master Project

Sep. 2025 – Present

University of Amsterdam

Supervisor: Prof. Jasper van Wezel & Prof. Jorik van de Groep

- Derived and implemented exciton-phonon scattering kernels from microscopic interaction Hamiltonians within a quantum kinetic framework, enabling numerical evaluation of momentum-resolved exciton population transfer.
- Developed transfer-matrix-method (TMM) modeling for planar multilayer stacks to analyze momentum-resolved exciton emission, radiative channels, and environmental modification of optical response.
- Built reproducible Python pipelines for simulating and fitting photoluminescence observables, including PL intensity and quantum yield, with automated parameter sweeps and visualization.

Dynamical Phase Transitions in the Kinouchi-Copelli Neuronal Network

Undergraduate Project

Sep. 2023 – Jan. 2024

National Tsing Hua University

Supervisor: Prof. Hsiu-Hau Lin & Prof. Fan Wang

- Analyzed a minimal U(1)-type neuron model and built its discrete Kinouchi-Copelli realization to study collective dynamics in fully-connected and lattice neuronal networks.
- Characterized two phase transitions via mean-field theory and numerical simulations: a quiescent-to-spontaneous asynchronous firing (SAF) transition with a critical branching ratio σ_c , and a SAF-to-spontaneous synchronous firing (SSF) transition featuring emergent synchronization and frequency-domain signatures.

HONORS & AWARDS

Second Prize, National Mathematics Competition for University Students (Provincial)

Mar 2023

Scholarship for Yan Ji-Ci Class, Institute of Physics, Chinese Academy of Sciences

2020–2022

First-Class Scholarship(Single Subject), Sichuan University

2019–2020

TECHNICAL SKILLS

Languages: Mandarin (Native), English (IELTS: 7.0)

Programming: Python, Mathematica, Git, L^AT_EX